

ECN 712 Fall 2025
EE Schlee
Problem Set VII, Due 30 October: Trick or Treat!

1. *Bidding in a Sealed-Bid Auction.* A person—bidder 1—is participating in a sealed-bid auction for an indivisible object with another bidder: the two bidders simultaneously and independently submit bids (nonnegative real numbers), and the bidder with the highest bid gets the object and pays its bid. Bidder 1's preferences for distributions over the object x and money m are represented by the vN-M utility $u(x, m) = xv_1 + m$, where $x \in \{0, 1\}$ and $0 < v_1 < 1$: that is, bidder 1's preferences are quasilinear (in some good other than this object) and bidder 1 is risk neutral. Bidder 1 is endowed with wealth of $w > 1$.

Suppose that, from the viewpoint of bidder 1, bidder 2's bid b_2 is uncertain, with distribution described by a C^1 cumulative distribution function F defined on $[0, 1]$ with density $f(b_2) = F'(b_2) > 0$ for all $b_2 \in [0, 1]$, $F(0) = 0$. (Note that ties have zero probability and so can be ignored.)

- (a) Write down the problem that determines bidder 1's choice of a bid, b_1 .
 - (b) Suppose that v_1 increases (but is still less than 1). What happens to bidder 1's bid? Explain.
 - (c) Suppose that bidder 1 becomes *strictly risk averse*: bidder 1's vN-M utility is now $T(u(x, m))$ where T is strictly increasing and *strictly* concave. What happens to bidder 1's bid? Prove it.
2. MWG, 6.E.2. (Wait, doesn't (c) look familiar? Do it again—better.)
3. A monopolist produces a good according to a constant returns to scale technology with average cost of $k > 0$ per unit. The demand $d(\cdot)$ is nonnegative, continuous, and decreasing on $\mathbb{R}_+$ with $d(k) > 0$; and on $\{p \geq 0 \mid d(p) > 0\}$, the demand is differentiable with $d' < 0$.
 - (a) What is the effect of an increase in k on the price and quantity? Prove it.
 - (b) Assume that d is C^2 . Find a sufficient condition for $p - k$ to be nondecreasing in $-k$ in the sense that if p_t solves the firm's problem at $k = k_t$ and $-k_1 > -k_0$, then $p_1 - k_1 \geq p_0 - k_0$. (Suggestion: set $y = p - k$.) For each of the following (commonly used) demands determine whether the price rises by *more than* or *less than* the unit-cost increase (that is, whether $p - k$ increases or decreases in k): (i) $d(p) = a - bp$; (ii) $\ln d(p) = a - bp$; and (iii) $d(p) = ap^{-b}$ for $b > 1$.¹ (If the good were produced by a competitive industry with all firms producing according to constant returns technology with average cost of k , then of course we would have $p = k$. What this problem illustrates is that an empirical finding of $dp/dk \approx 1$ does not imply that the industry is competitive.)²

¹The first two demands should really be written $d(p) = \max\{a - bp, 0\}$ and $\ln d(p) = \max\{a - bp, 0\}$, but you get the same solution by ignoring nonnegativity of quantity.

²Sumner, D., (1981) [Measurement of Monopoly Behavior: An Application to the Cigarette Industry, *The Journal of Political Economy*, 1010-1019] uses variations in cigarette tax rates across states to estimate dp/dk and finds it close to 1.

The next two questions are from my undergraduate Intermediate Microeconomics class and are meant to be a review.

4. An art museum operates according to the cost function $c(q) = F + kq$ for $q > 0$, with $c(0) = 0$, where c is measured in dollars, q denotes the number of visits to the museum. The inverse demand for museum visits is given by $p = a - bq$, where p is admission charge in dollars. The parameters a , b , F , and k are all positive with $a > k$.
 - (a) What can you say about the returns to scale of the museum's technology?
 - (b) Identify the values of F for which the museum's *maximum* profit is zero.
 - (c) *Evaluate*: "If the museum's profit is negative for every $q > 0$, then a benevolent social planner who is interested simply in total surplus—consumers' surplus plus profit—should not subsidize the museum to keep it open." A picture might help.

5. *Vertical Integration and Consumer Welfare*. A monopolist manufacturer sells its output to a monopolist retailer. The manufacturer chooses a wholesale price P to charge the retailer; the retailer, who takes that price as given, then chooses how many units to order from the manufacturer. Assume that the manufacturer's technology is constant returns to scale with an average cost of k , that the retailer's only costs are purchases from the manufacturer, and that the inverse demand for consumption of the good is given by $p = a - bq$. The parameters a , b , and k are all positive with $a > k$.
 - (a) Identify the price P set by the manufacturer, the quantity ordered by the retailer, and the final price to consumers. (Hint: start with the retailer's problem.)
 - (b) Suppose now that the two firms merge. Calculate the new consumer price and quantity. Are consumers better off with the merger? Explain intuitively why consumers might benefit from such a merger.

6. A monopolist sells a single good to N identical consumers, where N is a positive integer. If it sells $q \geq 0$ units to *each* consumer its profit is $Np(q)q - c(Nq)$, where $p(\cdot)$ is the inverse demand for each consumer and $c(\cdot)$ is the cost function. The functions $p(\cdot)$ and $c(\cdot)$ satisfy the usual conditions: $p(\cdot)$ is continuous and nonincreasing; and, on the set of outputs for which $p(\cdot)$ is positive, it is *strictly* decreasing; $c(0) = 0$; and c is differentiable with $c' > 0$.
 Besides these standard assumptions, the cost function $c(\cdot)$ is a *strictly concave*.³ Suppose that the size of the market, N , increases. Can you say whether per-consumer sales q go up or down? If you can, prove it. If not, can you find an additional condition that would allow to conclude how q changes with N ?

³What does this assumption imply about the firm's technology? Does it cohere with the fact that the good is produced by a monopoly?